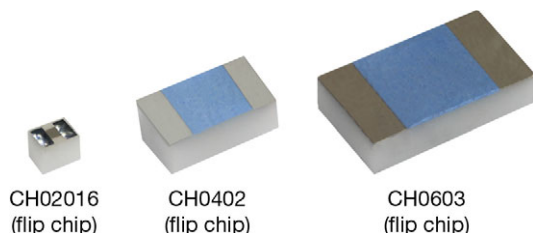


High Frequency 70 GHz Thin Film Chip Resistor


CH02016
(flip chip)

CH0402
(flip chip)

CH0603
(flip chip)

LINKS TO ADDITIONAL RESOURCES



3D Models



S-Parameters



Simulation Tools



Application Notes



Capabilities and Custom Options



Did You Know?



Infographics



Why It Matters

FEATURES

- Operating frequency 70 GHz
- Thin film microwave resistors
- Flip chip, wraparound or one face termination
- Ohmic range: 10 Ω to 500 Ω
- Design kits available
- Modelithics® library available
- Small internal reactance (LC down to 1×10^{-24})
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912



RoHS
COMPLIANT
HALOGEN
FREE
GREEN
(5-2008)

Those miniaturized components are designed in such a way that their internal reactance is very small. When correctly mounted and utilized, they function as almost pure resistors on a very large range of frequency, up to 50 GHz for CH0402 and CH0603, and 70 GHz for CH02016 from 10 Ω to 500 Ω .

STANDARD ELECTRICAL SPECIFICATIONS

MODEL	SIZE	RESISTANCE RANGE Ω	RATED POWER P_n W	LIMITING ELEMENT VOLTAGE V	TOLERANCE $\pm \%$	TEMPERATURE COEFFICIENT $\pm$ ppm/ $^{\circ}$ C
CH02016	02016	10 to < 50	0.100	30	5	100 (50 upon request)
CH02016	02016	50 to $\leq$ 500	0.100	30	2, 5	100 (50 upon request)
CH02016	02016	50 and 100	0.100	30	1, 2, 5	100 (50 upon request)
CH0402	0402	10 to < 50	0.300	37	2, 5	100 (50 upon request)
CH0402	0402	50 to $\leq$ 500	0.300	37	1, 2, 5	100 (50 upon request)
CH0603	0603	10 to < 50	0.400	50	2, 5	100 (50 upon request)
CH0603	0603	50 to $\leq$ 500	0.400	50	1, 2, 5	100 (50 upon request)

Note

(1) PCB mounting with +70 $^{\circ}$ C ambient temperature

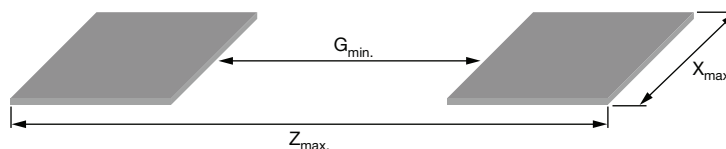
DIMENSIONS in millimeters (inches)

CH02016 F / CH02016 P / CH0402 P / CH0603 P 	CH0402 F / CH0603 F 	CH0402 N / CH0402 G / CH0603 N / CH0603 G
--	--------------------------------	--

CASE SIZE MODEL / TERMINATION	DIMENSIONS								
	A ± 0.10 (± 0.004)	B ± 0.10 (± 0.004)	C ± 0.127 (± 0.005)	D		E		F ± 0.050 (± 0.002)	G ± 0.050 (± 0.002)
CH02016 F CH02016 P	0.480 (0.020)	0.390 (0.016)	0.420 (0.016) (1)	MIN.	MAX.	MIN.	MAX.	± 0.050 (± 0.002)	± 0.050 (± 0.002)
CH0402 F				0.110 (0.004)	0.150 (0.006)	0.260 (0.010)	n/a	0.260 (0.010)	0.300 (0.012)
CH0402 N CH0402 G	1.000 (0.040)	0.600 (0.023)	0.500 (0.020)	0.150 (0.006)	0.350 (0.014)	0.150 (0.006)	0.350 (0.014)	n/a	n/a
				0.100 (0.004)					
CH0402 P	1.200 (0.047)	0.600 (0.023)	0.500 (0.020)	0.110 (0.004)	0.150 (0.006)	0.320 (0.013)	n/a	0.320 (0.013)	0.880 (0.035)
CH0603 F CH0603 N CH0603 G	1.520 (0.060)	0.850 (0.033)	0.500 (0.020)	0.250 (0.010)	0.510 (0.020)	n/a	n/a	n/a	n/a
CH0603 P	1.720 (0.068)	0.850 (0.033)	0.500 (0.020)	0.235 (0.009)	0.275 (0.011)	0.660 (0.026)	n/a	0.660 (0.026)	1.355 (0.053)

Note

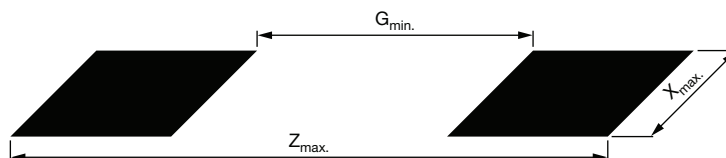
(1) ± 0.070 (± 0.003)

LAND PATTERN FOR F “FLIP CHIP” TERMINATIONS in millimeters (inches)


CHIP SIZE	$Z_{max.}$	$X_{max.}$	$G_{min.}$
02016	0.53 (0.021)	0.44 (0.017)	0.15 (0.006)
0402	1.40 (0.055)	0.65 (0.026)	0.40 (0.016)
0603	1.71 (0.067)	0.90 (0.035)	0.76 (0.030)

Note

- Suggested land pattern: according to IPC-7351

LAND PATTERN FOR N AND G WRAPAROUND TERMINATIONS in millimeters (inches)


CHIP SIZE	$Z_{max.}$	$X_{max.}$	$G_{min.}$
0402	1.55 (0.061)	0.73 (0.029)	0.15 (0.006)
0603	2.37 (0.093)	0.98 (0.039)	0.35 (0.014)

Dimension and tolerance of land pattern shall be defined by PCB designer; PCB can be designed according to IPC-7351A “Generic Requirements for Surface Mount Design and Land Pattern Standard”

PERFORMANCE (CH02016 F TERMINATION)
TEST PROCEDURES AND REQUIREMENTS

AEC-Q200 CLAUSE	TEST	PROCEDURE	GLOBAL PERFORMANCES	TYPICAL PERFORMANCES (25 Ω TO 250 Ω)
3	High temperature exposure	MIL-STD-202 method 108 1000 h at $T = 125^{\circ}\text{C}$, unpowered	$\pm 2\% \pm 0.05\ \Omega$	$\pm 0.2\% \pm 0.05\ \Omega$
4	Temperature cycling	JESD22 method JA-104 1000 cycles (-55°C to $+155^{\circ}\text{C}$)	$\pm 1.8\% \pm 0.05\ \Omega$	$\pm 1.5\% \pm 0.05\ \Omega$
7	Biased humidity	MIL-STD-202 method 103 1000 h $85^{\circ}\text{C} / 85\% \text{ RH}$ 10 % of operating power	$\pm 2\% \pm 0.05\ \Omega$	$\pm 0.75\% \pm 0.05\ \Omega$
8	Operational life	MIL-STD-202 method 108 condition D steady state $T = 125^{\circ}\text{C}$ at rated power 90' on / 30' off / 1000 h	$\pm 2.5\% \pm 0.05\ \Omega$	$\pm 1\% \pm 0.05\ \Omega$
13	Mechanical shock	MIL-STD-202 method 213 condition C 100 g/6 ms 3.75 m/s 3 shock/direction, 2 directions along 3 axes (18 shocks)	$\pm 0.05\% \pm 0.05\ \Omega$	$\pm 0.015\% \pm 0.05\ \Omega$
14	Vibration	MIL-STD-202 method 204 5 g for 20 min, 12 cycles each of 3 orientations Test from 10 Hz to 2000 Hz	$\pm 0.1\% \pm 0.05\ \Omega$	$\pm 0.05\% \pm 0.05\ \Omega$
15	Resistance to soldering heat	MIL-STD-202 method 210 condition D Flux used: alpha 611 Solder temp.: $260^{\circ}\text{C} \pm 5^{\circ}\text{C}$ Total immersion during 10 s	$\pm 2.5\% \pm 0.05\ \Omega$	$\pm 0.5\% \pm 0.05\ \Omega$

TEST PROCEDURES AND REQUIREMENTS				
AEC-Q200 CLAUSE	TEST	PROCEDURE	GLOBAL PERFORMANCES	TYPICAL PERFORMANCES (25 Ω TO 250 Ω)
17	ESD	AEC-Q200-002	Classification 1C 1000 V _{DC} to 2000 V _{DC}	
18	Solderability	J-STD-002 - Preconditioning 4 h dry heat aging and 235 °C SnPb 5 s - 215 °C SnPb 5 s - 260 °C SnAgCu 10 s	Good tinning (≥ 95 % covered) No visible damage	
20	Flammability	UL 94	Class V-0 No burning	
21	Board flex	AEC-Q200-005	± 0.1 % ± 0.05 Ω	± 0.05 % ± 0.05 Ω
24	Flame retardance	AEC-Q200-001	No flame, no explosion, no temperature higher than 350 °C	

PREFERRED MODELS AND VALUES

Vishay Sfernice highly recommend to use the smallest sizes and flip chip version to get the best performances.

Recommended Values:

10 Ω / 18 Ω / 25 Ω / 50 Ω / 75 Ω / 100 Ω / 150 Ω / 180 Ω /
200 Ω / 250 Ω / 330 Ω / 500 Ω

Those values are available with a **MOQ of 100 pieces**.

**Other values can be ordered upon request, but higher
MOQ will apply: 1000 pieces for CH02016, 500 pieces for
CH0402, 250 pieces for CH0603.**

Recommended termination:

F

Recommended tolerance:

2 %

DESIGN KITS

Design kits are available ex stock in CH02016 and CH0402 sizes. There are 20 pieces per recommended value. F termination.
5 % tolerance.

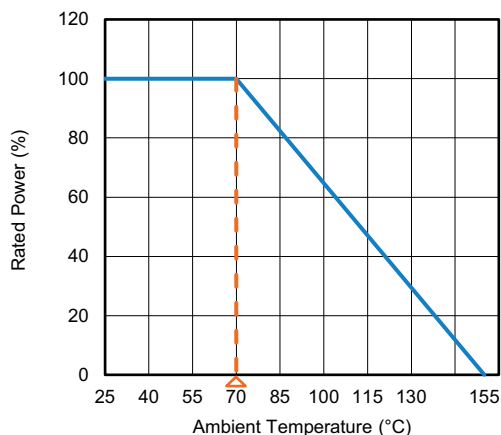
Those kits are packaged in pieces of tape and delivered in ESD bags.

TEST BOARDS

TRL (Thru Reflect Line) and DUT (Device Under Test) evaluation boards (50 Ω or 100 Ω) are available on request.

THERMAL SPECIFICATIONS (CH0402 AND CH0603)

POWER DERATING CURVE



PCB MOUNTING	
CHIP SIZE	RATED POWER ⁽¹⁾ (W)
0402	0.300
0603	0.400

Note

⁽¹⁾ PCB mounting with +70 °C ambient temperature.

PCB is FR4 base material 100 mm x 150 mm x 1.5 mm, 35 μm
Cu-layer

**SUGGESTIONS TO INCREASE POWER**

PCB MOUNTING AMBIENT +25 °C			
CHIP SIZE	POWER (W)		
	FR4 ⁽¹⁾	ALUMINA ⁽²⁾	AIN ⁽²⁾
0402	0.450	1.000	1.200
0603	0.580	1.820	2.220

Notes

- (1) Thermal measurement with Optris Xi 400 camera. Test performed with standard FR4 (PCB thickness 1.6 mm, copper thickness 35 µm)
- (2) Estimations with PCB thermal resistance knowledge

BACKSIDE CHIP TEMPERATURE +25 °C	
CHIP SIZE	POWER ⁽¹⁾ (W)
0402	2.000
0603	5.000

Note

- (1) Estimations with components thermal resistance knowledge

PACKAGING

Standard packaging is plastic tape and reel for all sizes.

Paper tape and reel is available for sizes 0402 and 0603 with F, N, and G terminations.

Depending on the type of terminations, parts will be packed differently:

One face:

- Gold terminations: (P termination option): Active face up.
Please use M termination code for active face down in tape and reel.
- Tin / silver terminations: (F termination option): Active face down in tape and reel.

Notes

- CH02016 with active face down in tape and reel have back-side blue marked to indicate right orientation
- Please refer to Vishay Sfernice Application Note [“Guidelines for Vishay Sfernice Resistive and Inductive Components”](#) for soldering recommendation (document number 52029, section “3. Guidelines for Surface Mounting Components (SMD)”, profile number 3 applies

SIZE	MOQ	NUMBER OF PIECES PER PACKAGE		TAPE WIDTH
		TAPE AND REEL		
		MIN.	MAX.	
02016	See MOQ mentioned on preferred models and values.	100	10 000	8 mm
0402			5000	
0603				

PACKAGING RULES**Tape and Reel**

See Part Numbering information to get the quantity desired by tape.

In regard to the CH02016 size only, up to 5 empty cavities can be found every 1000 parts in the reel. Nevertheless, the number of requested parts will be respected.

**GLOBAL PART NUMBER INFORMATION**

New Global Part Numbering: CH0402-50RJF (preferred part number format)

C	H	0	4	0	2	-	5	0	R	J	F	T		9	9	9
GLOBAL MODEL		SIZE		OHMIC VALUE		TOLERANCE		TERMINATION				PACKAGING		OPTION		
CH		02016 0402 0603		10R to 500R		F = 1 % G = 2 % J = 5 %		F (flip chip): SnAg over nickel barrier N (W/A): SnAg over nickel barrier (except 02016) P (one face): ⁽¹⁾ gold bonding pads M (one face): gold bonding pads with active face down in tape and reel G (W/A): gold over nickel barrier (except 02016)				For more information see Codification of Packaging table		From 1 to 3 digits. Leave blank if no option.		

Historical Part Number Example: CH02016-100RGFPT1K (tapes of 1000 pieces)
CH0402-50RJF (waffle pack)

CHKIT Part Numbers ⁽²⁾: CHKIT-02016
CHKIT-0402

Notes

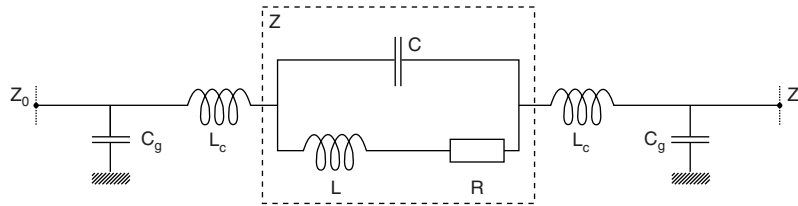
- Historical part numbers are not recommended but can still be used for ordering
- (1) Gold termination for application in hermetic package. Can also be mounted on PCB with SnAg solder paste. Please use M termination code for active face down in tape and reel
- (2) CHKIT for 0603 size is not available

CODIFICATION OF PACKAGING**SAMPLES-CUT TAPE (available for all sizes)**

CT	20 min., 20 mult.; 20 pcs max.
----	--------------------------------

PLASTIC TAPE (available for all sizes / TG for 02016 only)

T	100 min., 100 mult.; delivered in reels of 5000 pcs max. (TA to TG preferred)
TA	100 min., 100 mult.; delivered in reels of 100 pcs
TB	250 min., 250 mult.; delivered in reels of 250 pcs
TC	500 min., 500 mult.; delivered in reels of 500 pcs
TD	1000 min., 1000 mult.; delivered in reels of 1000 pcs
TF	5000 min., 5000 mult.; delivered in reels of 5000 pcs
TG	10 000 min., 10 000 mult.; delivered in reels of 10 000 pcs (02016 only)

TYPICAL HIGH FREQUENCY PERFORMANCE ELECTRICAL MODEL


C	Internal shunt capacitance
L	Internal inductance
R	Resistance
Z	Internal impedance (R, L, C)
L _c	External connection inductance
C _g	External capacitance to ground

The complex impedance of the chip resistor is given by the following equations:

$$Z = \frac{R + j\omega(L - R^2C - L^2C\omega^2)}{1 + C[(R^2C - 2L)\omega^2 + L^2C\omega^4]}$$

$$\frac{|Z|}{R} = \frac{1}{1 + C[(R^2C - 2L)\omega^2 + L^2C\omega^4]} \times \sqrt{1 + \left[\frac{\omega(L - R^2C - L^2C\omega^2)}{R} \right]^2}$$

$$\theta = \tan^{-1} \frac{\omega(L - R^2C - L^2C\omega^2)}{R}$$

Notes

- $\omega = 2 \times \pi \times f$
- f : frequency

R, L and C are relevant to the chip resistor itself.

L_c and C_g also depend on the way the chip resistor is mounted.

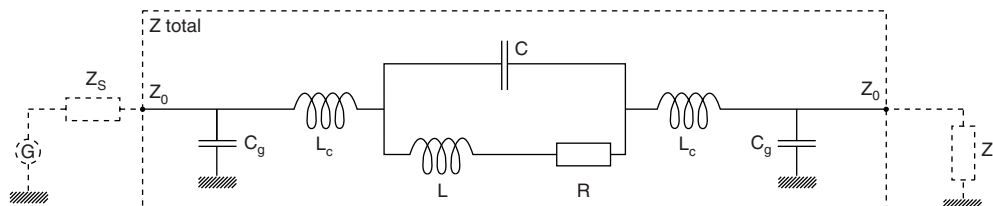
It is important to notice that after assembly the external reactance of L_c and C_g will be combined to internal reactance of L and C. This combination can upgrade or downgrade the HF behavior of the component.

This is why we are displaying three sets of data:

- $\frac{|Z|}{R}$ versus frequency curves which aim to show at a glance the intrinsic HF performance of a given chip resistor
- $\frac{|Z_{total}|}{R}$ versus frequency curves which aim to show the behavior of the chip resistor when mounted

These lines are terminated with adapted source and load impedance respectively Z_s and Z_l with Z₀ = Z_L = Z_s = R (for others configurations please consult us).

Equivalent circuit for S-parameters:

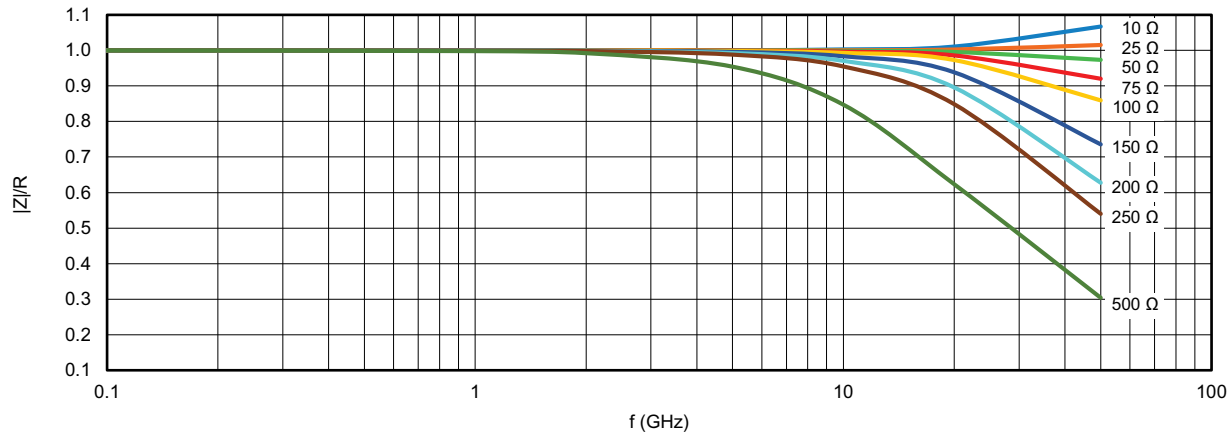


S-parameters are computed taking into account all the resistive, inductive and capacitive elements (Z total) and Z₀ = Z_L = Z_s = R.

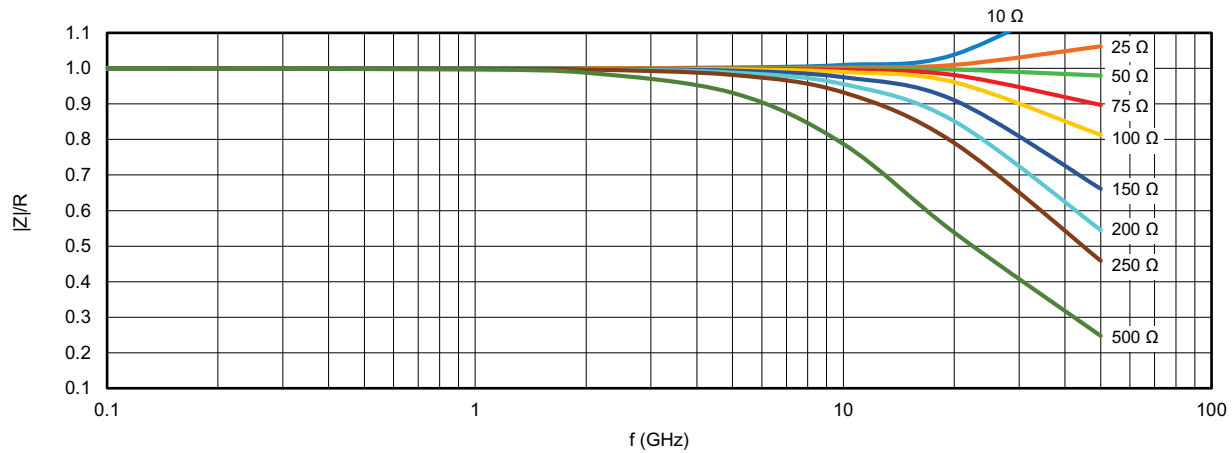
For simulation purposes, those S-parameter data are available for download here: www.vishay.com/doc?53061



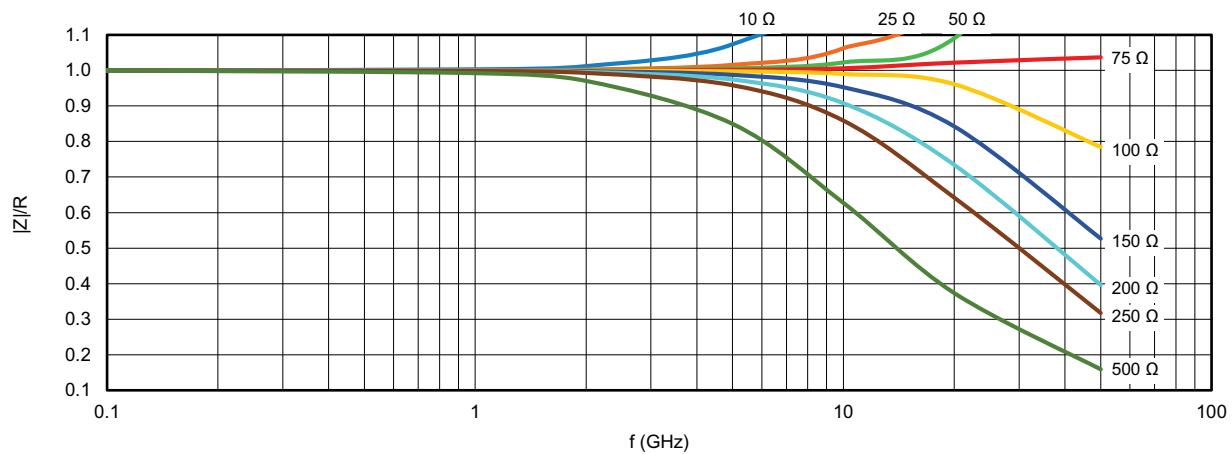
INTERNAL IMPEDANCE CURVES



Internal impedance curve for 02016 size (F and P terminations)



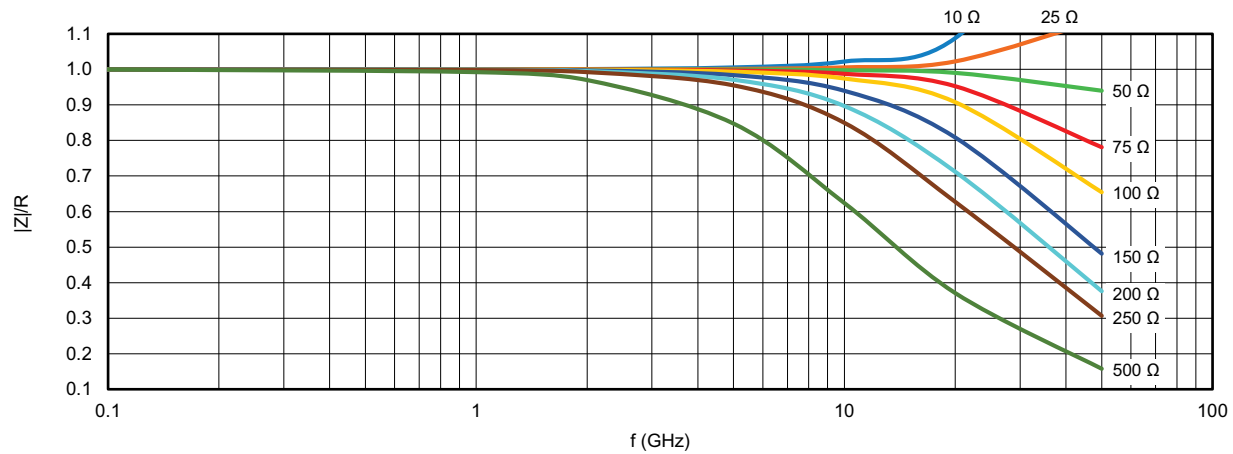
Internal impedance curve for 0402 size (F and P terminations)



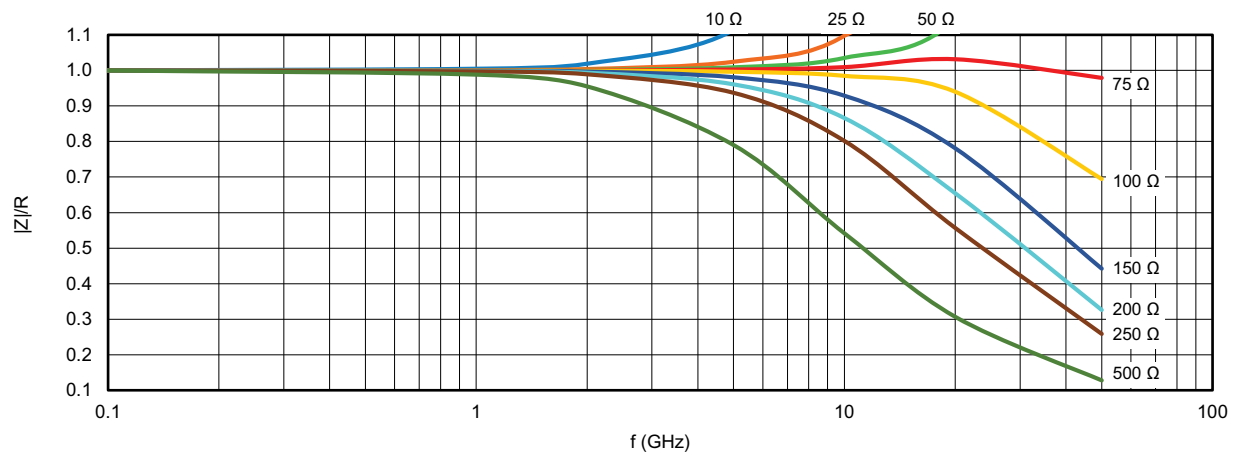
Internal impedance curve for 0402 size (N and G terminations)



INTERNAL IMPEDANCE CURVES



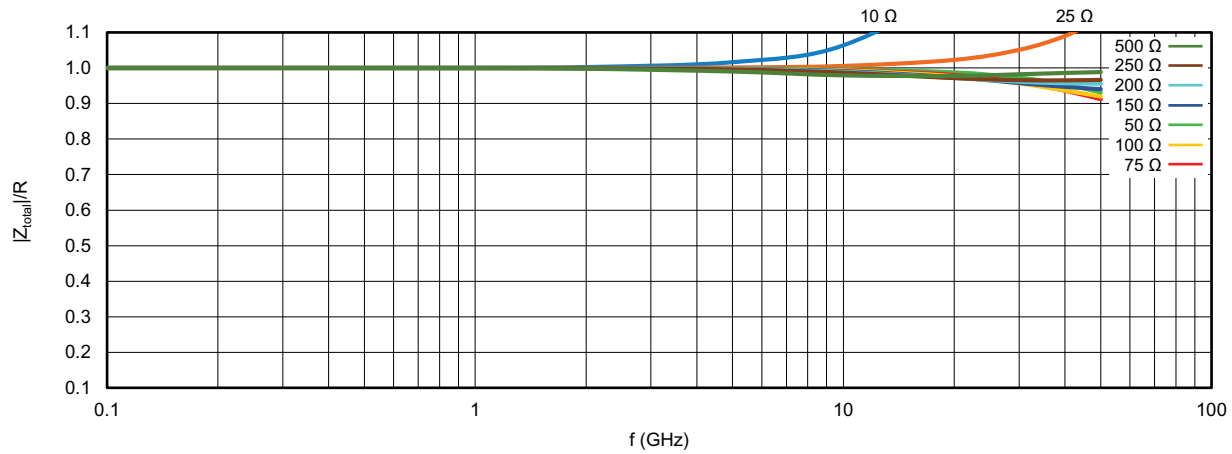
Internal impedance curve for 0603 size (F and P terminations)



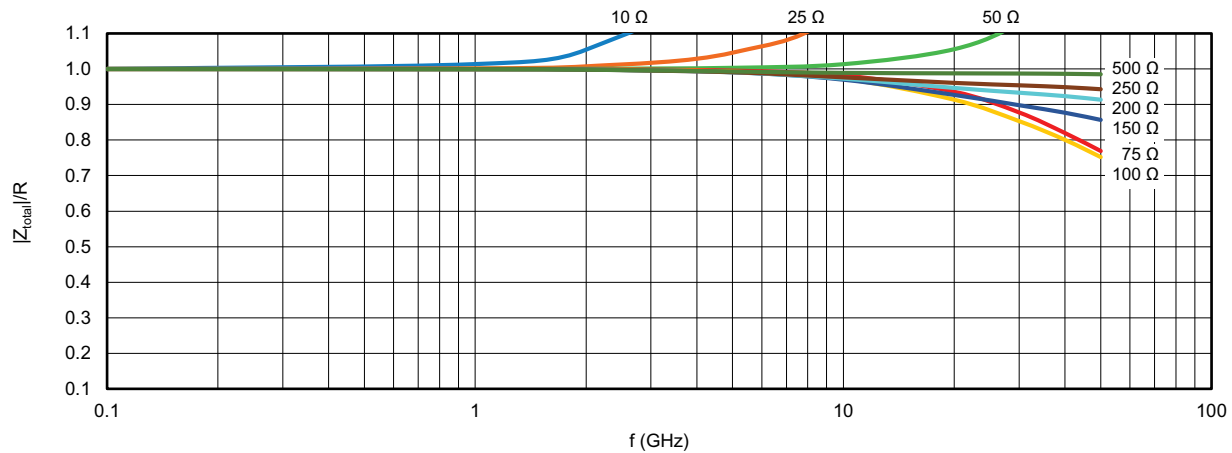
Internal impedance curve for 0603 size (N and G terminations)



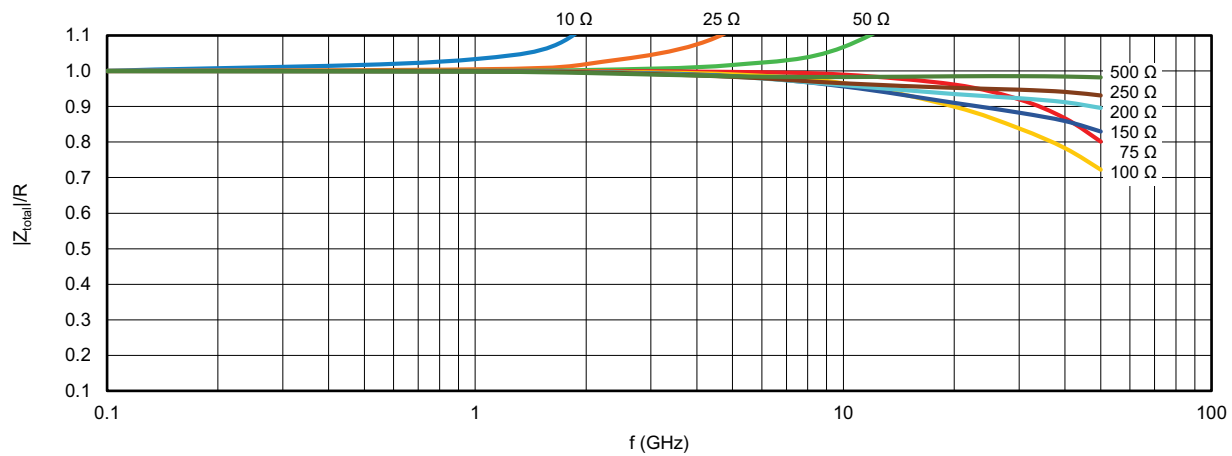
INTERNAL IMPEDANCE CURVES ($|Z_{TOTAL}| / R$)



Internal impedance curve for 02016 size (F and P terminations)



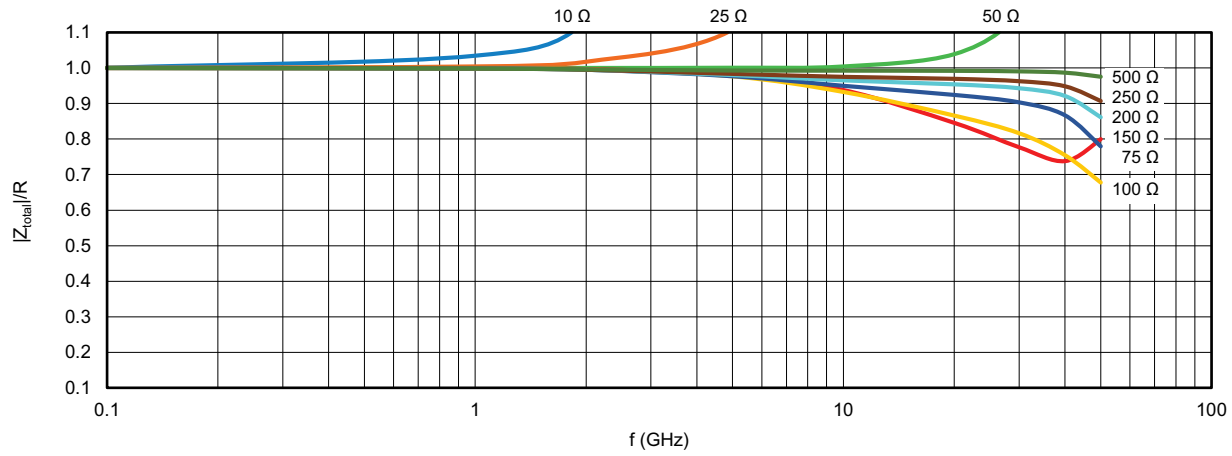
Internal impedance curve for 0402 size (F and P terminations)



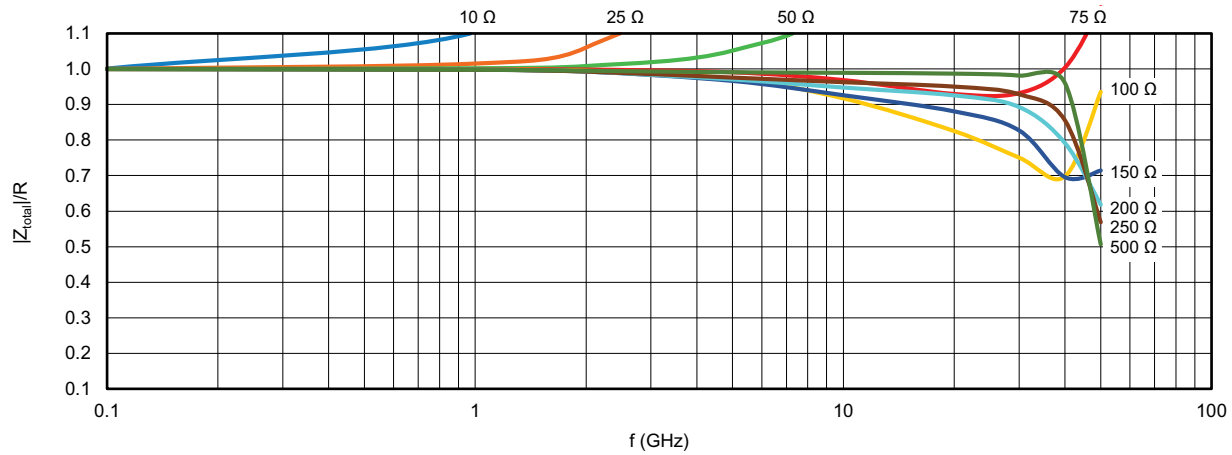
Internal impedance curve for 0402 size (N and G terminations)



INTERNAL IMPEDANCE CURVES ($|Z_{TOTAL}| / R$)



Internal impedance curve for 0603 size (F and P terminations)

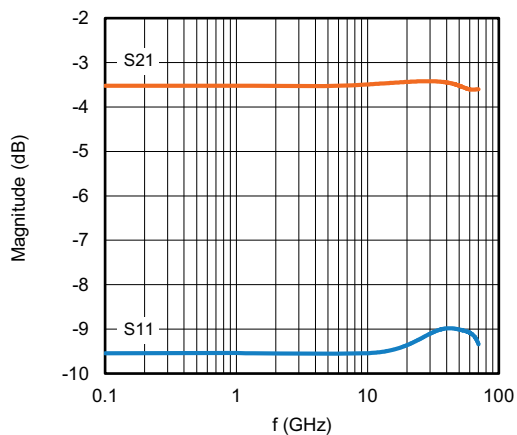


Internal impedance curve for 0603 size (N and G terminations)

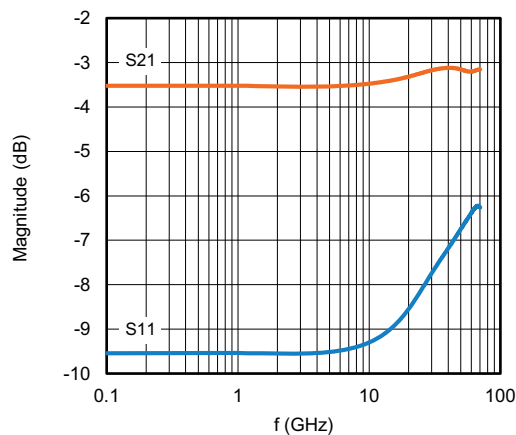


S-PARAMETER

CH02016 (F and P Terminations)

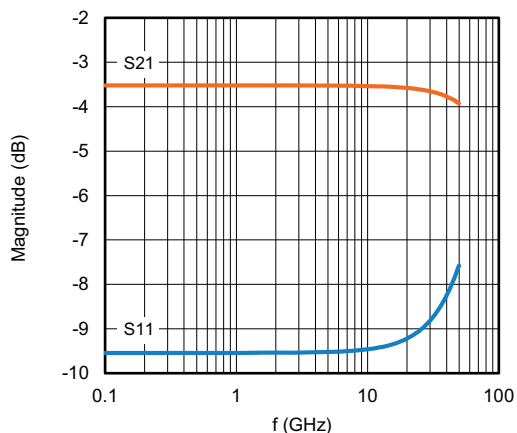


CH02016 flip chip ($Z_0 = Z_l = Z_s = R = 50 \Omega$)

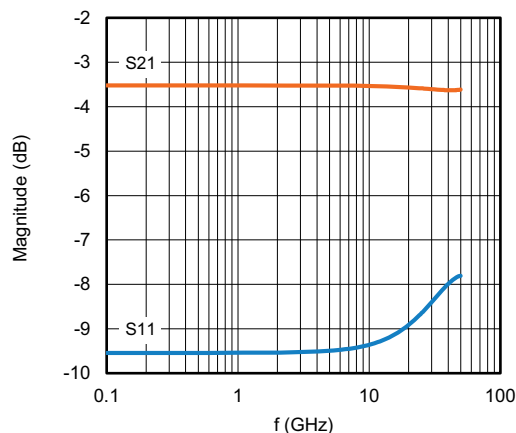


CH02016 flip chip ($Z_0 = Z_l = Z_s = R = 100 \Omega$)

CH0402 (F and P Terminations)

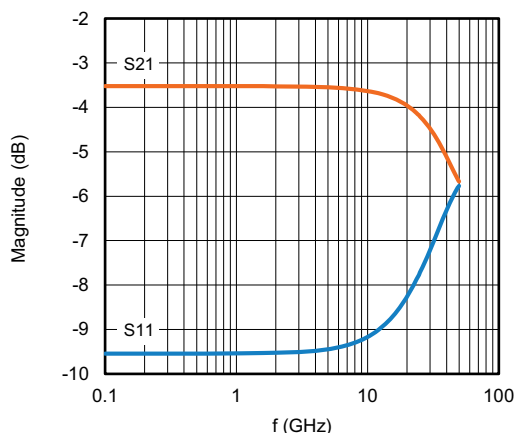


CH0402 flip chip ($Z_0 = Z_l = Z_s = R = 50 \Omega$)

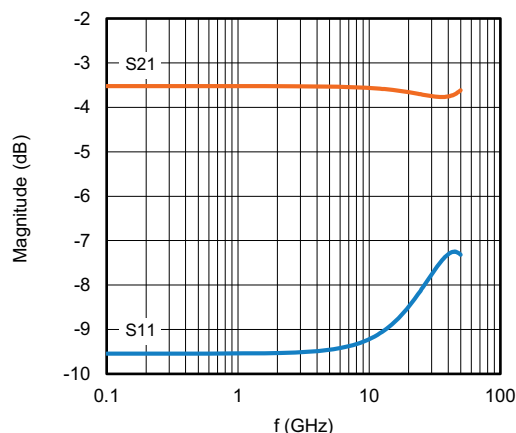


CH0402 flip chip ($Z_0 = Z_l = Z_s = R = 100 \Omega$)

CH0402 (N and G Terminations)



CH0402 wraparound ($Z_0 = Z_l = Z_s = R = 50 \Omega$)

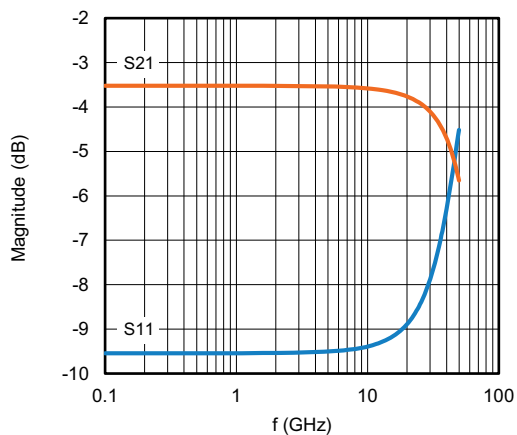


CH0402 wraparound ($Z_0 = Z_l = Z_s = R = 100 \Omega$)

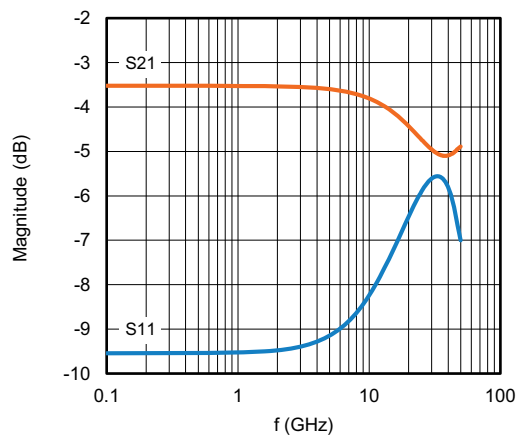


S-PARAMETER

CH0603 (F and P Terminations)

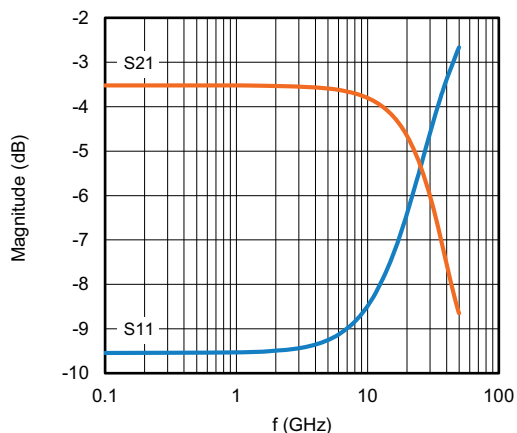


CH0603 flip chip ($Z_0 = Z_1 = Z_s = R = 50 \Omega$)

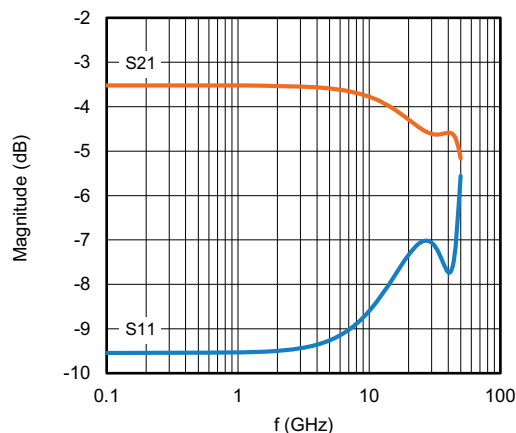


CH0603 flip chip ($Z_0 = Z_1 = Z_s = R = 100 \Omega$)

CH0603 (N and G Terminations)



CH0603 wraparound ($Z_0 = Z_1 = Z_s = R = 50 \Omega$)



CH0603 wraparound ($Z_0 = Z_1 = Z_s = R = 100 \Omega$)



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